FOR A VALUATION RING OF DIMENSION > 1, R[[X]] IS FLAT OVER R BUT NOT FLAT OVER R[X]

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The opening sentence 'The arguments above show that this is true if we can show that R is not completely normal' relies on the context span; the reduction and the almost-integral construction of 1/y are both inside the counted span. Flatness over R itself comes from the context paragraph (valuation rings coherent), completing the flat-over-R / not-flat-over-R[x] pair.

Editorial note. The label lemma-completion-polynomial-ring-not-flat-bis names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 1042-1083: Badam-Baplan construction - a valuation ring R is coherent (Algebra, example-valuation-ring-coherent) so R[[x]] is flat over R (Algebra, proposition-characterize-coherent); and the same-section lemma proving: if R[[x]] is flat over R[x] for a domain R with fraction field K, then R is normal iff completely normal (almost-integral elements detected via (ax-b)f = -rb and the faithfully flat localization at S = {h : h(0)=1})
- Valuation rings are normal (Algebra, lemma-valuation-ring-normal); almost integral elements: ralpha^n in R for all n

Source reference. Original definition 00GW: sources/stacks-chow/tags.txt, source lines 622–622. Prerequisite pointer invoked at 1117; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0EWV: sources/stacks-chow/tags.txt, source lines 18635–18635. Prerequisite pointer invoked at 1107; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00HQ: sources/stacks-chow/tags.txt, source lines 652–652. Prerequisite pointer invoked at 1129; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00IC: sources/stacks-chow/tags.txt, source lines 674–674. Prerequisite pointer invoked at 1094; it names a standard fact of a sibling

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chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 05CZ: sources/stacks-chow/tags.txt, source lines 6937–6937. Prerequisite pointer invoked at 1109; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

It turns out there is a ring R such that $R[[x]]$ is flat over R, but $R[[x]]$ is not flat over $R[x]$. See this post by Badam Baplan. Namely, let R be a valuation ring. Then R is coherent (Algebra, Example 0EWV) and hence $R[[x]]$ is flat over R by Algebra, Proposition 05CZ. On the other hand, we have the following lemma.

Lemma 0.1. *Let R be a domain with fraction field K. If $R[[x]]$ is flat over $R[x]$, then R is normal if and only if R is completely normal (Algebra, Definition 00GW).*

Proof. Suppose we have $\alpha \in K$ and a nonzero $r \in R$ such that $r\alpha^n \in R$ for all $n \geq 1$. Then we consider $f = \sum r\alpha^{n-1}x^n$ in $R[[x]]$. Write $\alpha = a/b$ for $a, b \in R$ with b nonzero. Then we see that $(ax - b)f = -rb$. It follows that rb is in the ideal $(ax - b)R[[x]]$. Let $S = \{h \in R[x] : h(0) = 1\}$. This is a multiplicative subset and flatness of $R[x] \to R[[x]]$ implies that $S^{-1}R[x] \to R[[x]]$ is faithfully flat (details omitted; hint: use Algebra, Lemma 00HQ). Hence

$$S^{-1}R[x]/(ax - b)S^{-1}R[x] \to R[[x]]/(ax - b)R[[x]]$$

is injective. We conclude that $hrb = (ax-b)g$ for some $h \in S$ and $g \in R[x]$. Writing $h = 1 + h_1x + \ldots + h_dx^d$ shows that we obtain

$$1 + h_1x + \ldots + h_dx^d = (1/r)(\alpha x - 1)g$$

This factorization in $K[x]$ gives a corresponding factorization in $K[x^{-1}]$ which shows that α is the root of a monic polynomial with coefficients in R as desired. □

Lemma 0.2. *If R is a valuation ring of dimension > 1, then $R[[x]]$ is flat over R but not flat over $R[x]$.*

Proof. The arguments above show that this is true if we can show that R is not completely normal (valuation rings are normal, see Algebra, Lemma 00IC). Let $\mathfrak{p} \subset \mathfrak{m} \subset R$ be a chain of primes. Pick nonzero $x \in \mathfrak{p}$ and $y \in \mathfrak{m} \setminus \mathfrak{p}$. Then $xy^{-n} \in R$ for all $n \geq 1$ (if not then $y^n/x \in R$ which is absurd because $y \notin \mathfrak{p}$). Hence $1/y$ is almost integral over R but not in R. □